Grade: Check Plus

Assignment 2

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Question 1: Mean and Standard Deviation of Stock Price ✓ Q1

$E(X) = \Sigma\, x \cdot f(x) = \$25.41$

$Var(X) = E(X^2) - [E(X)]^2 = 653.17 - (25.41)^2 = 653.17 - 645.6681 = 7.5019$

$\sigma = \sqrt{7.5019} \approx \$2.739$

Question 2: Probabilities Within Standard Deviations

a. Within 1 standard deviation: $\mu \pm \sigma = 25.41 \pm 2.739 \rightarrow (22.67, 28.15)$

✓ Q2a

Values of X in this range: 23, 24, 25, 26, 27, 28

$P = .08 + .09 + .12 + .15 + .13 + .10 = 0.67$

b. Within 2 standard deviations: $\mu \pm 2\sigma = 25.41 \pm 5.478 \rightarrow (19.93, 30.89)$ ✓ Q2b

$P = 1.00$

Question 3: Distribution of Option Value

The option allows purchase at \$25 (the strike price). The option value $V = \max(X - 25, 0)$.

Stock Price (x)	Option Value (v)	f(x)
20	0	.05
21	0	.06
22	0	.07
23	0	.08
24	0	.09
25	0	.12

Stock Price (x)	Option Value (v)	f(x)
26	1	.15
27	2	.13
28	3	.10
29	4	.10
30	5	.05

✓ Q3

Consolidated distribution of V:

v	P(V = v)
0	.47
1	.15
2	.13
3	.10
4	.10
5	.05
Total	1.00

Question 4: Probability Option Has Positive Value

✓ Q4

The option has positive value when the stock price exceeds \$25 (i.e., $x > 25$):

$P(V > 0) = P(X = 26) + P(X = 27) + P(X = 28) + P(X = 29) + P(X = 30)$

$P = .15 + .13 + .10 + .10 + .05 = 0.53$

Question 5: Mean and Standard Deviation of Option Value

Total 1.00 1.36 4.42

$E(V) = \$1.36$

$Var(V) = E(V^2) - [E(V)]^2 = 4.42 - (1.36)^2 = 4.42 - 1.8496 = 2.5704$

✓ Q5

$\sigma_V = \sqrt{2.5704} \approx \$1.603$